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Dual-Channel Deepfake Audio Detection: Leveraging Direct and Reverberant Waveforms

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Abstract

Synthetic voice generation algorithms typically model direct acoustic paths but fail to accurately reproduce subtle physical acoustic reverberation patterns present in real-world recorded environments. This paper introduces a dual-channel deep learning model that isolates direct path sound from room reverberant tail waveforms.

Key Methodology & Architecture

Blind Room Impulse Response (RIR) estimation, dual-stream CNN-LSTM network, phase-coherence cross-correlation.

Datasets & Benchmark Environments

ACE Challenge Dataset, LibriSpeech RIR Synthesized Corpus, Real Office/Hall Recordings.

Performance Metrics & Graph Axis Parameters

Reported Results: Deepfake Detection Accuracy = 97.8%, AUC-ROC = 0.992, F1-Score = 0.976.

Graph Parameter Mapping: X-axis: Room Reverberation Time (RT60 in seconds, 0.1s to 1.2s); Y-axis: Deepfake Detection Accuracy (80% to 100%).

Comparative Analysis

Advantages (Pros)	Limitations (Cons)
Leverages physical acoustic properties that generative AI cannot synthesize naturally.	Performance degrades slightly in near-anechoic recording studios.

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